

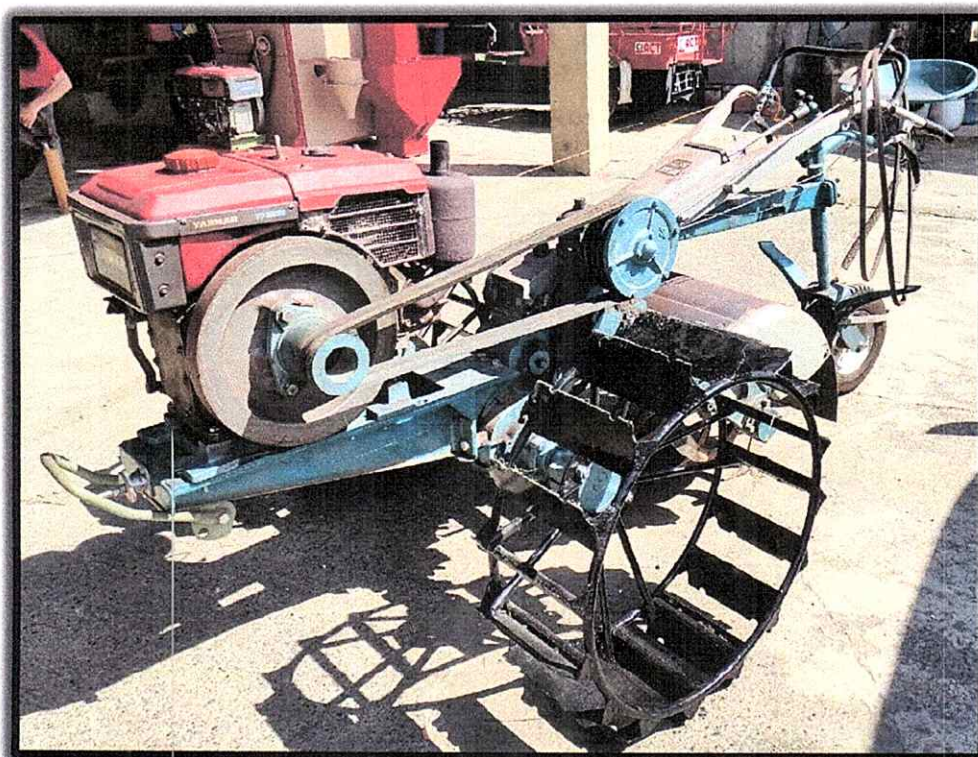


AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0286



Agricultural Machinery – Walking-Type Agricultural Tractor NEW LION RK125

Test Report Number	2023 – 0286
Date of Issuance	July 08, 2024
Valid Until	July 08, 2029
Agricultural Machinery	Walking-Type Agricultural Tractor
Type	Rotary tilling, Axle-driven
Brand	NEW LION
Model	RK125
Test Requested by	ACT (MACHINERIES AND METAL CRAFT) CORPORATION Brgy. Turayong, Cauayan City, Isabela 3305

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS 346:2022	AMTEC Test
Power requirement of the prime mover, kW, maximum	13.5	8.95
Actual Field Capacity, ha/h	ND ^a	0.094
Peak Transmission Efficiency, %, minimum	80	NM ^b
Breakdowns/malfunctions during the five-hour continuous running test.	Shall have none	ND ^b

^a Not specified by the test applicant.

^b The laboratory testing of the machine was not conducted as decided by the test applicant.

Note: ND – No Data, NM – Not Measured

OBSERVATIONS

1. Operator's manual	Not provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	NEW LION Vietnam
4. Serial number	No data
5. Prime mover used	8.95 kW YANMAR TF120M Diesel Engine
6. Number of operators	One
7. Handling and stability of the WTAT	The machine was stable and could be maneuvered without issues when working on a straight path and when turning at corners during the field test operation.
8. Manipulating of the operating levers	The operating levers are accessible to the operator.
9. Replacing and adjusting of parts	The belt tension can be adjusted by moving the engine along its base.
10. Safety features	The rotary tilling blades of the WTAT are provided with a cover.
11. Failure or abnormalities that may be observed on the WTAT or its component parts.	No failure or abnormalities were observed during the field testing of the WTAT.
12. The WTAT shall be generally made of steel bars and sheet metals.	The WTAT is generally made of 1100 Series Low Alloy Steel (LAS) as determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer. Refer to Table 1 for the relative elemental composition of the materials.
13. The chain and sprocket used shall conform to PAES 303:2000 (Engineering materials – Roller chains and sprockets for agricultural machines – Specifications and applications) and shall have provisions for refilling of lubricating oil.	There is a provision for refilling the lubrication oil for the chain and sprocket transmission system. However, the conformity of the chain and sprocket assembly to PAES 303:2000 was not verified due to the inaccessibility of the components.
14. The hand grip shall be made of non-slip material and shall have a minimum nominal diameter of 25 mm.	The hand grip is made of rubber and has a diameter of 35.16 mm.
15. Mechanism for handle bar height adjustment with at least three settings shall be provided. The handle bar height shall be 650 mm to 800 mm.	The height of the handlebar is adjustable by placing the bolt attachment in different settings. The handlebar height was set at 1170 mm high from the ground.

OBSERVATIONS (Cont'n)

16. The speed control lever shall be provided and accessible to and operated from the perspective of the operator. It shall be provided with a detent mechanism capable of setting and retaining/locking the speed at any point within the range of the engine speed.	The vertical-type speed control lever is located at the operator's front and center. A detent mechanism is provided.
17. The main clutch lever shall be provided and accessible to and operated from the perspective of the operator.	The main clutch lever is vertical-type and is located on the left-hand side of the operator.
18. Stand assembly for use when the WTAT is not in operation shall be provided.	The WTAT is provided with a stand assembly.
19. There shall be earmuffs or other ear protection devices provided for the operator to use when 95 dB(A) is exceeded during operation. Recommendations for use of ear protection device shall likewise be indicated in the operator's manual.	The maximum noise level measured during the test was 90.8 dB(A). Earmuffs or other ear protection devices were not provided during the field performance testing of the WTAT.
20. The WTAT shall be free from any manufacturing defects that may be detrimental to its operation.	There were no observed manufacturing defects on the WTAT.
21. All metal surfaces shall be painted properly.	All metal surfaces of the WTAT are painted properly.
22. Parts of the WTAT that are exposed to the operator shall be free from sharp edges and rough surfaces.	The exposed parts of the WTAT are free from sharp edges and rough surfaces.
23. Appropriate labels, safety symbols, and warning notices shall be provided in accordance with PNS/BAFS 330:2022 (Technical means for ensuring safety – Guidelines).	Not provided
24. All controls shall be in accordance with PNS/BAFS 330:2022 (Technical means for ensuring safety – Guidelines).	The controls are accessible and can be controlled without issues by the operator under normal working conditions. However, the controls do not have labels specifying their function.
25. All exposed, hot surfaces shall be provided with heat shield or protection to minimize the possibility of inadvertent contact with the operator.	Not provided
26. Mechanism for emergency stop of the engine and immediate load disengagement of the transmission system shall be provided.	The load can be immediately disengaged from the transmission system by releasing the clutch lever.
27. Mechanism for belt transmission adjustment shall be provided.	The belt tension can be adjusted by moving the engine along its base.

OBSERVATIONS (Cont'n)

28. Belt cover and mud guard shall be provided. All other moving parts shall have safety guards. When the PTO is in use, a cover or casing that protects the sides of the PTO shaft shall be fitted. An additional non-rotating casing shall also be provided when the PTO is not in use. This casing shall enclose the PTO shaft completely and shall be fixed to the tractor body.	The machine is provided with a rubber mud guard and a cover for the input shaft pulley.
29. There shall be a provision for means to minimize vibration.	It was not verified if the machine's vibration was minimized/reduced due to limited technical capabilities to quantify and compare vibration levels during the test operation.
30. The hexagonal axle shall be in accordance with the specifications of PAES 108:2000 (Agricultural machinery – Hexagonal axle and hub for walking-type agricultural tractor – Specifications).	A circular axle is provided for both the wheel and tilling blade axles. The diameter of the wheel and tilling blade axles are 40.38 mm and 50.26 mm, respectively.
31. Other remarks	a) The laboratory test was not conducted as decided by the test applicant. b) Cage wheels were used during the test operation. c) Nameplate and warning notices were not available at the time of test.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Dates of Test	June 8, 2023
Location of Test	Brgy. Cabaruan, Cauayan City, Isabela 16°56'32.60" N, 121°44'9.64" E
Items Tested	Depth of Rototilling, Width of Rototilling, Field Capacity, Field Efficiency, Noise Level, Number of Operators, and Fuel Consumption

METHODS OF TEST

The rotary-tilling type WTAT was tested based on the PNS/BAFS 348:2022 Agricultural Machinery – Walking-type Agricultural Tractor – Methods of Test.

DESCRIPTION OF THE MACHINE

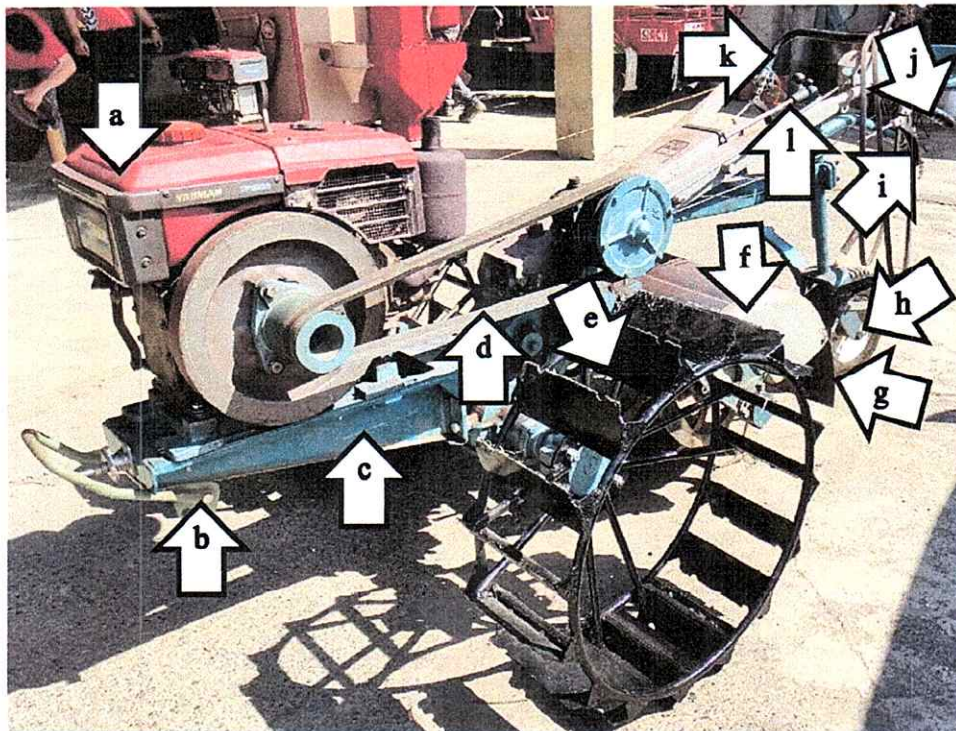


Figure 1. Parts of the NEW LION RK125 Walking-Type Agricultural Tractor powered by 8.95 kW YANMAR TF120M Diesel Engine include the following: (a) Prime mover, (b) Stand assembly, (c) Chassis, (d) Belt and pulley power transmission system, (e) Cage wheel, (f) Rotary tilling blades assembly, (g) Mud guard, (h) Depth wheel, (i) Steering clutch lever, (j) Handle bar, (k) Main clutch lever, and (l) Gear shift lever.

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification
1 Main structure		
1.1 Overall dimensions, mm		
1.1.1 Length	ND	2740
1.1.2 Width	ND	1480
1.1.3 Height	ND	1300
1.1.4 Ground clearance	ND	110
1.2 Weight, kg	ND	NM ^b
2 Engine		
2.1 Brand	YANMAR	YANMAR
2.2 Model	TF120M	TF120M
2.3 Make or manufacturer	YANMAR CO., LTD. Indonesia	YANMAR CO., LTD. Indonesia
2.4 Serial number	32038	32038
2.5 Type	ND	Four-stroke cycle, single horizontal cylinder
2.6 Rated power, kW	8.95	8.95 ^a
2.7 Rated speed, rpm	2400	2400 ^a
2.8 Displacement, cm ³	ND	ND ^c
2.9 Cooling system	ND	Water-cooled
2.10 Starting system	ND	Manual cranking lever
3 Type of clutch		
3.1 Main clutch	ND	Lever-operated (Dog clutch)
3.2 Steering clutch	ND	Handle-operated (Dog clutch)
3.3 Tilling clutch	ND	Lever-operated
4 Power transmission system		
4.1 Prime mover to input	ND	V-belt and pulley
4.2 Prime mover ^d	ND	110 × 3B × NM ^e
4.3 Input ^d	ND	215 × 3B × NM ^e
4.4 Belt size	B-1930 Li	B-1930 Li

^a The data were obtained from the prime mover nameplate.

^b There is no available instrument to facilitate the measurement of the parameter during the test operation.

^c Not indicated on the prime mover nameplate.

^d Pulley diameter, mm × number of belts × shaft diameter, mm

^e The component part is inaccessible for measurement.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
5	Axle		
5.1	Type	ND	Circular
5.2	Overall length of shaft, mm	ND	NM ^b
5.3	Width across the flat, mm	ND	NA
5.4	Diameter, mm	ND	40.38
6	Hexagonal hub	ND	NA
7	Rotary tilling wheel		
7.1	Number of blades	ND	18
7.2	Number of flanges	ND	NA
7.3	Blade thickness, mm	ND	9.06
7.4	Blade type	ND	C-shape
7.5	Material	ND	1100 Series LAS ^c
7.6	Operating width, mm	ND	565
8	Safety devices	ND	Rotary tilling blade cover, Mud guard
9	Special features	ND	None

^a The test applicant did not provide complete machine specifications.

^b The component part cannot be disassembled for measurement at the time of test.

^c Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer; see Table 1 for the elemental composition of the material.

Note: NA – Not Applicable, NM – Not Measured

RESULTS

Item		Data
1	Test conditions	
1.1	Condition of field	
1.1.1	Type	Lowland
1.1.2	Location	Brgy. Cabaruan, Cauayan City, Isabela
1.1.3	Soil type	Clay loam
1.1.4	Dimensions of field	
1.1.4.1	Length, m	32.0
1.1.4.2	Width, m	16.0
1.1.4.3	Area, m ²	512
1.1.5	Depth of water, mm (for lowland)	34.89
1.1.6	Number of days soaked (for lowland)	ND ^a
1.1.7	Soil resistance, kg/cm ²	0.5
1.1.8	Spacing of stubbles, rows × hills, mm	NA
1.1.9	Height of stubbles, mm	388
1.1.10	Weed density	High
1.2	Condition of the tractor	
1.2.1	Implement/Attachment used	
1.2.1.1	Type	Rotary tiller
1.2.1.2	Operating width, mm	565
2	Field performance	
2.1	Type of field operation	Primary tillage – Rotary tilling
2.2	Field operational pattern	Circuitous
2.3	Traveling speed, kph	2.06
2.4	Average depth of rotary tilling, mm	109
2.5	Average width of rotary tilling, mm	566

^a Not specified by the test applicant.

Note: NA – Not Applicable, ND – No Data

RESULTS (Cont'n)

Item		Data
2.6	Actual field capacity	
2.6.1	ha/h	0.094
2.6.2	h/ha	10.68
2.7	Theoretical field capacity	
2.7.1	ha/h	0.117
2.7.2	h/ha	8.58
2.8	Field efficiency, %	80.34
2.9	Fuel consumption	
2.9.1	L/h	1.50
2.9.2	L/ha	16.07
2.10	Noise at operator's ear level, dB(A)	90.8
2.11	Unrototilled, %	11.72

RESULTS (Cont'n)

Table 1. Elemental composition of the rotary tilling blades of the NEW LION RK125 Walking-Type Agricultural Tractor.

Element	Percent Composition
Vanadium	0.010
Chromium	0.020
Manganese	0.400
Iron	99.520
Cobalt	<LOD
Nickel	<LOD
Copper	0.020
Niobium	<LOD
Molybdenum	0.020
Tin	<LOD
Tungsten	<LOD
Grade	1100 Series Low Alloy Steel

^a Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

OBSERVATIONS



Figure 2. Quality of work.



Figure 3. Markings on the NEW LION RK125 Walking-Type Agricultural Tractor.

Tested by:


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Verified by:


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Approved for Release:


ARTHUR L. FAJARDO

AMTEC Director

JUL 11 2023

Date

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